WORKED SOLUTION

STEP I 2000, Question 2

Pure • Binomial Expansion Coefficients

STEP I 2000, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q2

The key idea

Expanding two binomials with fractional terms and then hunting for matching powers is messy. It is much cleaner to pull out the negative powers of x first, so that each bracket becomes a polynomial in a single power of x (x^6 or x^2). Then the second product turns out to be the first one in disguise, thanks to the factorisation $x^6 - 1 = (x^2 - 1)(x^4 + x^2 + 1)$, so the "hence" is almost free.

Solution

The first expansion

Taking out the lowest power of x from each bracket,

$$\left(x^4 - \frac{1}{x^2}\right)^5 = \left(\frac{x^6 - 1}{x^2}\right)^5 = x^{-10}\,(x^6 - 1)^5,$$
$$\left(x - \frac{1}{x}\right)^6 = \left(\frac{x^2 - 1}{x}\right)^6 = x^{-6}\,(x^2 - 1)^6.$$

So the whole expression is $x^{-16}P(x)$, where

$$P(x) = (x^6 - 1)^5\,(x^2 - 1)^6.$$

The coefficient of x^n in the original expansion is therefore the coefficient of x^{n+16} in the polynomial $P(x)$.

By the binomial theorem,

$$(x^6 - 1)^5 = \sum_{j=0}^{5} \binom{5}{j} (-1)^{5-j} x^{6j}, \qquad (x^2 - 1)^6 = \sum_{k=0}^{6} \binom{6}{k} (-1)^{6-k} x^{2k},$$

so the coefficient of x^m in $P(x)$ is the sum of

$$\binom{5}{j} (-1)^{5-j} \binom{6}{k} (-1)^{6-k}$$

over all pairs with $6j + 2k = m$, $0 \le j \le 5$, $0 \le k \le 6$.

Coefficient of x^{-12}. We need $m = -12 + 16 = 4$, that is $6j + 2k = 4$. Since $6j \le 4$ forces $j = 0$, the only pair is $(j, k) = (0, 2)$, giving

$$\binom{5}{0}(-1)^5 \cdot \binom{6}{2}(-1)^4 = (-1)(15) = -15,$$

as required.

Coefficient of x^2. Now $m = 2 + 16 = 18$, so $6j + 2k = 18$, i.e. $3j + k = 9$ with $k \le 6$. That allows $j = 1, 2, 3$:

(j,k)	from $(x^6-1)^5$	from $(x^2-1)^6$	product
$(3,0)$	$\binom{5}{3}(-1)^2 = 10$	$\binom{6}{0}(-1)^6 = 1$	10
$(2,3)$	$\binom{5}{2}(-1)^3 = -10$	$\binom{6}{3}(-1)^3 = -20$	200
$(1,6)$	$\binom{5}{1}(-1)^4 = 5$	$\binom{6}{6}(-1)^0 = 1$	5

So the coefficient of x^2 is $10 + 200 + 5 = 215$.

? **Ask yourself** Why can't $j = 0$ or $j = 4$ contribute to x^{18}?

With $j = 0$ we would need $k = 9$, and with $j = 4$ we would need $k = -3$; neither lies in $0 \le k \le 6$. It is easy to miss a pair (or invent one) here, so it pays to list the possibilities systematically.

The second expansion

? **Ask yourself** What has $x^4 + x^2 + 1$ got to do with the first part?

It is the other factor of $x^6 - 1$ when we treat it as a difference of cubes in x^2:

$$x^6 - 1 = (x^2)^3 - 1 = (x^2 - 1)(x^4 + x^2 + 1).$$

Splitting $(x^2 - 1)^{11}$ as $(x^2 - 1)^5 (x^2 - 1)^6$,

$$(x^2-1)^{11}(x^4+x^2+1)^5 = [(x^2-1)(x^4+x^2+1)]^5 (x^2-1)^6 = (x^6-1)^5(x^2-1)^6 = P(x).$$

So the second expansion is exactly $P(x)$, which is x^{16} times the first expansion.

Coefficient of x^4. This is the coefficient of x^4 in $P(x)$, which we found above (it corresponded to x^{-12} in the first expansion). It is -15.

Coefficient of x^{38}. This corresponds to $6j + 2k = 38$, i.e. $3j + k = 19$ with $j \leq 5$, $k \leq 6$. Since $k \leq 6$ we need $3j \geq 13$, so $j = 5$ and $k = 4$, giving

$$\binom{5}{5}(-1)^0 \cdot \binom{6}{4}(-1)^2 = 15.$$

So the coefficient of x^{38} is 15.

There is a neat way to see this without counting. P has degree $30 + 12 = 42$, and replacing x by $1/x$ gives

$$x^{42}P(1/x) = (1 - x^6)^5(1 - x^2)^6 = (-1)^{11}P(x) = -P(x).$$

Comparing coefficients, the coefficient of x^{42-m} in P is minus the coefficient of x^m. With $m = 4$, the coefficient of x^{38} is $-(-15) = 15$, which agrees.

Checks

- Put $x = 1$ in $P(x)$: it vanishes, so all the coefficients of P add up to 0. That's a quick sanity check if you expand fully.
- We multiplied out both products exactly by computer (as lists of coefficients). We found -15 for x^{-12} and 215 for x^2 in the first, and -15 for x^4 and 15 for x^{38} in the second, and we confirmed that the second product has degree 42 and equals x^{16} times the first.
- Evaluating the original expression at random rational values of x matches the computed expansion exactly.

Common slips

- Sign errors: with x^{6j} and x^{2k} as the chosen powers, the sign of a term is $(-1)^{5-j}(-1)^{6-k} = -(-1)^{j+k}$, not $(-1)^{j+k}$. Using the wrong one flips every answer.
- Forgetting the $(1, 6)$ or $(3, 0)$ terms in the x^2 coefficient, and getting 200 or 205.
- Expanding $(x^2 - 1)^{11}(x^4 + x^2 + 1)^5$ directly without spotting the factorisation: possible, but long and error-prone.

Beyond STEP

Writing a product of sums as a single sum and picking out one power of x is the idea behind *generating functions*. A coefficient of a product counts the ways to make a total from the parts, exactly as we counted pairs (j, k) above. The symmetry $x^{42}P(1/x) = -P(x)$ says that P is an (anti)*palindromic* polynomial; such polynomials turn up in the study of cyclotomic polynomials and roots of unity, of which $x^6 - 1$ is a first example. For more problems in this spirit, Siklos's *Advanced Problems in Mathematics* is a good next step, and Allenby's *Rings, Fields and Groups* covers polynomial factorisation properly.

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